Confidence scoring, evidence tiers, and state-space compilation

Confidence Scoring and Evidence Tiers

The shipped `ConfidenceModel` in `../cogant/py/cogant/translate/confidence.py` computes a scalar score in $[0, 1]$ from:

- Average confidence over provenance records.
- An evidence-diversity term (scaled, capped).
- A parser certainty factor applied multiplicatively.
- Conflict penalties subtracted after scaling.

$$c = \max\left(0,\ \min\left(1,\ (\bar{e} + \delta_d) \cdot \kappa - \pi\right)\right) \tag{1}$$

Here $\bar{e}$ is the mean evidence confidence, δ_d is the diversity bonus (bounded), κ is parser certainty, and π aggregates conflict penalties. **Tiers** (for example static-only versus static-plus-runtime) are assigned from score thresholds and evidence source tags (`determine_confidence_tier`); see the same module for named thresholds and enum values. The manuscript does not duplicate those literals so they cannot drift from code.

State space and behavior

Where traces or coverage are available, **dynamic** extraction feeds the state-space compiler. The goal is a compact behavioral model: states, actions, transitions, and observations that sit alongside the static graph for tasks that require execution-sensitive features. The tuple (S, A, T, O) of variables, actions, transitions, and observation modalities mirrors the structure of a partially observed Markov decision process as used in discrete active inference [@parr2022active; @dacosta2020active; @smith2022stepbystep], and at the level of reachable state/transition graphs it also resembles the Kripke structures traditionally used in model checking [@clarke1999model], without attempting to discharge temporal-logic obligations. PyMDP [@heins2022pymdp] is one such downstream consumer that executes the compiled state-spaces as discrete active inference simulations over the exported Generalized Notation Notation bundles.

The shipped `StateSpaceCompiler` in `../cogant/py/cogant/statespace/compiler.py` constructs this model in several coordinated passes driven by the semantic